LESSON PLAN

LOW SPEED FLIGHT (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Wheels types of wheels and description of typical main wheel, general repair to wheels.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Wheels types of wheels and description of typical main wheel, general repair to wheels.
5. Specific Objectives	9.1.1.1 Trainee will be able to describe various Aircraft engineering systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of wheel components on a typical main wheel assembly; Safety: Ensure aircraft is chocked and brakes are set; Supervision: Instructor verifies component identification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft wheel assembly?
(b) Name two common types of aircraft wheels used in engineering.
(c) Why is the main wheel critical compared to the nose wheel?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Classification of aircraft wheel types • Description of typical main wheel assembly • Identification of main wheel components • Function of wheel bearing and hub • Role of brake assembly in wheel system	L & Ds	PPT, IB	15 min
(b)	• General repair procedures for wheel assemblies • Inspection criteria for wheel damage • Handling of wheel bearing units • Torque specifications for wheel nuts • Documentation of wheel maintenance actions	L & Ds	PPT, PDF	20 min
(c)	• Common defects in wheel structures • Repair limitations and replacement criteria • Safety checks post-repair verification • Storage requirements for removed wheels • Reporting defects via engineering log	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a main wheel and a nose wheel in terms of load bearing.
(b) List three critical inspection points when performing general repair to wheels.
(c) Describe the procedure for verifying torque specifications on wheel nuts after installation.

12. Home Assignment and Follow Up	Prepare a brief note on the variation of tyre pressure with all-up weight as covered in the next lesson plan (LP 06), highlighting the relationship between weight and pressure.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Tyres and tubes, variation of tyre pressure with all up weight, creep and creep marking on wheels.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min